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Enhancing Intrusion Detection with CNN-RF Hybrid Model: A High-Performance Approach Using NSL-KDD Dataset

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Abstract—**The growing sophistication of attacks is accompanied by the growing significance of having developed sophisticated intrusion detection systems (IDS) that can accurately determine the threats. This paper thus suggests improving the identification of network intrusions by combining RF with CNN. The model depends on the hierarchical feature extraction by CNN for the classification and ensemble learning of RF for the selection of features. In terms of the experimental results obtained, the proposed model beats traditional IDS strategies when tested on the NSL-KDD dataset with a recall of 0.96, an F1-score of 0.965, an accuracy of 98.73%, and a precision of 0.97. The use of this more complex data preprocessing method provides a guarantee that it would be able to detect unique types of attacks and is more resilient to imbalanced datasets. The technology is adaptable as well as scalable, showcasing the potential to be developed and applied for real-time intrusion detection.**

Keywords—Intrusion Detection System, Random Forest, Convolutional Neural Network, NSL-KDD, Cybersecurity, Network Security, Feature Extraction, Hybrid Model.

I. Introduction

The rising number of network intrusions has made it essential to create strong security systems that protect information and maintain the integrity of digital infrastructures [1]. Network intrusion detection systems are a must to identify and prevent malicious activities in a network. Classical intrusion detection systems, signature- or anomaly-based, is unreliable in the changing dynamics of networks [2]. The signature-based methodologies fail to address unique, previously unknown threats since they rely on pre-existing attack signatures [3]. Anomaly-based methods often suffer from a large number of false positives and are thus not useful for large-scale, dynamic networks [4]. These constraints call for better detection algorithms which will naturally be able to cope with emergent attack vectors. In the past decades, optimism has certainly been generated that deep learning techniques, particularly ANNs, could fine-tune the capabilities of IDS in terms of adaptability and accuracy. However, deep learning has produced the most promising results in intrusion detection so far; nonetheless, there are still problems related to optimization of network designs, imbalanced dataset management, and detection performance in real-time applications [5]. Datasets such as KDD99 and NSL-KDD may be historically important but cannot document the dynamic nature of cyber threats or complexities of modern-day network traffic [6]. Computational effectiveness is often ignored in research when the focus deals with deployment in resource-constrained environments, such as embedded devices, and edge networks. The reasons for this work are based on an interest to advance in current IDS. Its goal is to gain insight into deep feed-forward networks and other kinds of artificial neural networks. This proposed system would significantly improve detection performance by integrating a new data preprocessing pipeline and enhanced neural network model. Complex feature selection approaches eliminate the difficulties that come with analyzing high-dimensional network data [7]. A hybrid loss function would also be incorporated to cater to both known and unknown attacks. Another feature is its ability to identify zero-day assaults better by dynamically adapting to emerging patterns of assault instead of using fixed signatures. Even during heavy network traffic, the real-time data processing infrastructure of the system also assures minimal latency. The proposed system is tested and evaluated with various attack scenarios across all different types of network environments to ensure that it is scalable and resilient. The integration of these enhancements into intrusion detection systems is aimed at filling the gaps identified in the literature and developing more scalable, adaptive systems that can be highly effective. Fig. 1 depicts the IDS Classification model tree.

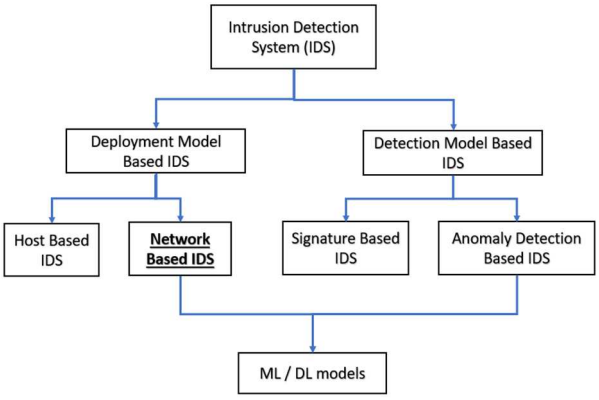

Fig. 1. IDS Classification Model

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II. Related Works

Network intrusion detection systems have become increasingly important for securing dynamic digital infrastructures from cyber threats. In their work, Rele and Patil [8] proposed a hybrid NIDS to increase accuracy and reduce false positives by combining techniques related to ML, DL, and anomaly detection. In a similar direction, Dalal and Rele [9] presented an ML-based architecture that predicts cybersecurity malware in sandbox environments with the focus of being proactive in threat protection. Although the authors do highlight the promising features of ML and DL to enhance the identification of risks, their work primarily revolves around classification tasks. Sophisticated DL topologies have also been explored as part of NIDS systems. Li [11] proposed a model to fine-tune an intrusion detection system specifically to address imbalanced datasets and operated using CNN-BiLSTM architecture that enhanced the performance of detection by 6.7%. Chen et al. [14] also did studies indicating the fact that training models using unprocessed network data rather than using a feature-engineered model produces better performance. Such findings depict the ability of deep learning in controlling complex dynamics of networks. Despite their virtues, these models are unable to accommodate certain real-world class differences and computational complexity. Kiran et al. [10] stress the need for high IDS due to the increasing reliance on the internet. The authors present some evidence that machine learning can enhance the IDS detection capabilities so that attacks not according to pattern may also be discovered. In the work by Sivakami et al. [12], intrusions were detected on hospital networks using a Copula GAN-boosted Random Forest and samples from the minority classes have been synthesized to overcome the problem of class imbalances. The enhanced classification accuracy provided by this approach is extremely useful for all industries, particularly the healthcare sector. Finally, Deng and Qiao [13] further advanced the concept by discussing the development of NIDSs and proposing neural network architecture-based systems for real-time detection. However, because they can only be applied to static datasets, they are useless in real-time networks. Lonare et al. [15] and other comprehensive evaluations classify NIDS as hybrid, signature, or anomaly. They do so by evaluating the relative merits of approaches such as scalable K-means and MCF-MVO-ANN thereby showing the merits of DL. Yang et al. [16] proposed a hierarchical proxy-based design to enhance security in niche network contexts; besides it equips wireless mesh networks with an intrusion detection system. Although significant progress has been made, several gaps still exist. Most improvements in detection accuracy have been based on neglecting optimization strategies for deployment in resource-constrained environments. Many works have addressed class imbalances but lacked scalability evaluations in large-scale networks. Domain-specific models also require evaluation in broader contexts. Standardized benchmarks for hybrid models are also largely missing, preventing a comparative evaluation between studies. Further research is required for filling these gaps in order to design an efficient and scalable NIDS solution. Fig. 2 illustrates that among the various statistical techniques, publications pertaining to intrusion detection using the time-series statistical approach had the greatest citation values and publication count [17]. Fig. 3 shows that the value of publication count and citations based on expert systems is higher than that of finite state machines (FSM) and descriptive language. .Fig. 4 illustrates, published works on neural network based intrusion detection are more numerous than those on SVM and fuzzy logic.

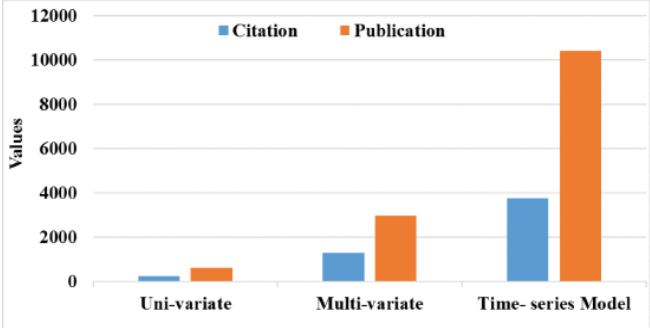

Fig. 2. Statistical Based Approach graph

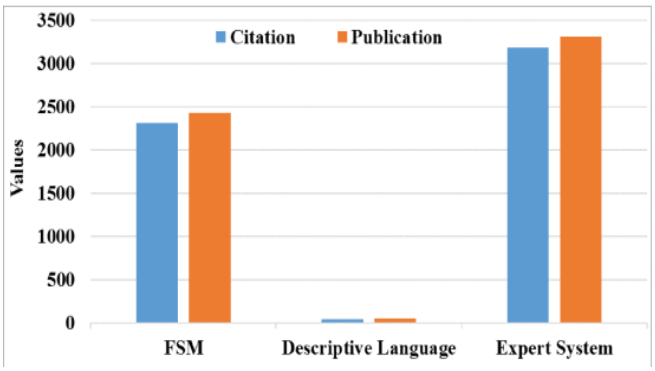

Fig. 3. Knowledge Based Approach graph

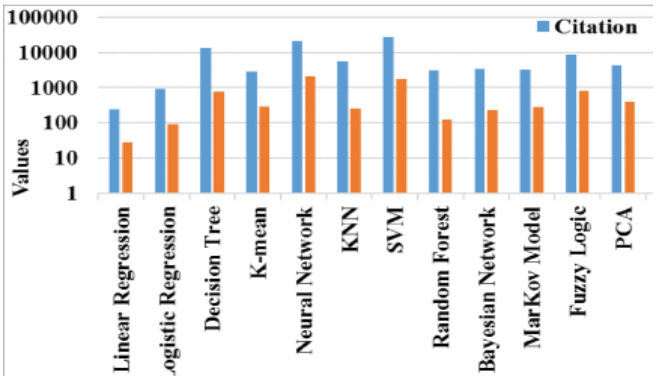

Fig. 4. ML Based Approach graph

III. Methodology

The proposed method enhances the efficiency of an IDS involves integrating state-of-the-art ML and DL algorithms. The steps for the system include preprocessing, feature extraction, model training, and classification. Each step has been carefully designed to address a specific challenge arising from network intrusion detection, which guarantees optimal performance in the identification of malicious network traffic. The predefined synergistic functioning of the proposed system will use the strengths of the different models to deliver scalable intrusion detection with accuracy and efficiency. Fig. 5 shows the flow graph of the system.

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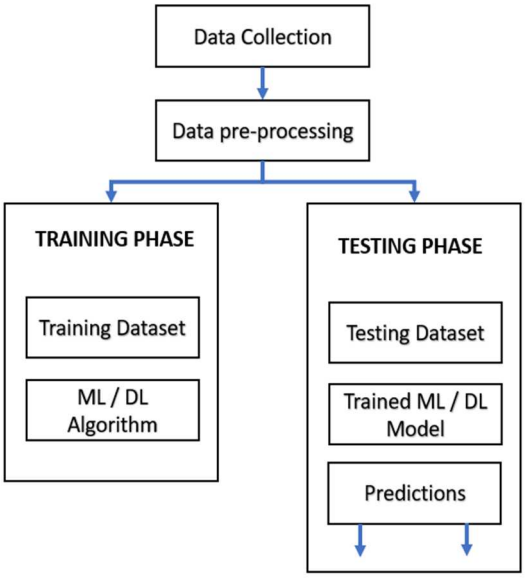

Fig. 5. System Flow Graph

A. Preprocessing and Feature Extraction

Preprocessing the data from the network is a necessary step for conducting successful analysis.The NSL-KDD dataset was chosen because it can handle imbalanced and redundant problems associated with the KDDCup99 dataset. The dataset is divided into three classes that feature forty-one attributes: fundamental, content-based, and traffic-based. The three features involve both numerical and categorical data describing the types of protocol used, service ports, and other connection identifiers. There are five classes of data involved in it. The pre-processing step of category variables into numerical forms takes place using encoding, feature scaling, and normalization. The formula for scaling numerical characteristics between 0 and 1 through min-max normalization. The formula for min-max is $x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$. It denotes the initial value as x, minimum and maximum values for the feature as x_{min} and x_{max}respectively. The latter guarantee allows this network to converge during training. Label encoding replaces all numerical labels into numerical representations while balancing techniques like Synthetic Minority Oversampling Technique (SMOTE), are implemented to handle imbalances in classes. The choice of features is crucial for dimensionality reduction of the given dataset in retaining critical features. This system combines to tackle the problems using Recursive Feature Elimination (RFE) and Principal Component Analysis (PCA). The formula is used to maximize the variance to transform the dataset into an orthogonal feature space via PCAis $Y = XW$. The input feature matrix is represented by Xwhile the transformation matrix is represented by$W$, and the transformed feature vector is represented by Y.This method can possibly lead to performance in computation augmented with reduced redundancy. While building the trees in the Random Forest model, every feature has a corresponding weight which is, then, what defines the feature significance scores. The ratings are used by RFE to iteratively remove features that are insignificant. PCA and RFE are combined to train the model by utilizing the most relevant features.

B. Model Architecture

A system is developed with a hybrid architecture that makes use of Random Forest (RF) for preliminary categorization and Convolutional Neural Networks (CNN) for deep feature learning. Firstly, the RF uses its ensemble-based decision-making procedure as a baseline model to classify objects. Bagging techniques are utilized by the RF algorithm to produce many decision trees. A single class is then produced by each tree as a prediction. The outcome of the forecast is determined through a consensus vote, as depicted below in (1).

$$p(y|X) = \frac{1}{T}\sum_{t=1}^{T} P_t(y|X) \quad (1)$$

The prediction probability of the tth tree, the combination of the predictions, and the number of trees are represented by $P_t(y|X)$and $p(y|X)$ respectively. The random sampling approach of RF is very effective for reducing overfitting.This is done by training a CNN with RF feature extraction. CNNs consist of convolutional, pooling, and fully connected layers. Filters are used in the convolutional processes is calculated as shown in (2).

$$Y[i,j] = \sum_m \sum_n X[i+m, j+n]W[m,n] + b \quad (2)$$

Here, the input matrix is denoted by X, the filter kernel is denoted by W, and the bias is denoted by b. The spatial dimensions can be estimated as: The spatial dimensions are reduced by applying max pooling using equation (3).

$$P[i,j] = \max(X[i{:}\, i+f, j{:}\, j+f]) \quad (3)$$

The size of the pooling window is denoted by$f$. A fully connected layer uses softmax activation to classify multiple classes after the output coming from the CNN, which is calculated using (4).

$$P(y_i|X) = \frac{\exp(z_i)}{\sum_j \exp(z_j)} \quad (4)$$

Where z_i stands for the output logits for class$i$. CNNs excel at detecting intricate patterns in network traffic, which is very much necessary to detect extra entry points for attacks.A hybrid system is developed by combining RF with CNN, which combines the best properties of ensemble learning and deep learning.At the same time, RF reduces dataset noise and rapidly and accurately classifies datasets to a baseline that is easily comprehensible.Deep hierarchical features are extracted from the RF filtered output using a CNN. The ultimate choice is determined by the outputs of CNN, which are more effective at identifying intricate attack patterns. The resulting flow guarantees a very reliable pipeline for detection, combining the computational efficiency of RF and the accuracy of CNN.The process of training and optimizing models A combination of hyperparameter optimization and supervised learning is used to train the model. Grid Search and cross-validation are used to tune the parameters of the RF, which include: the number of trees, depth, and the minimum samples per split. Bayesian Optimization is used on CNN hyperparameters. Training is above all focused on lowering the loss associated with categorical cross-entropy. The training objective minimizes the categorical cross-entropy loss, expressed in (5).

$$L = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} y_{ij}\log(p_{ij}) \quad (5)$$

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Where N represents the number of samples, C represents the number of classes, y_{ij} represents the actual value, and p_{ij} represents the probability of the result. The Adam optimizer adjusts the learning rate in accordance with the first and second moments of the gradients during optimization. Fig. 6 depicts the proposed system flow diagram.

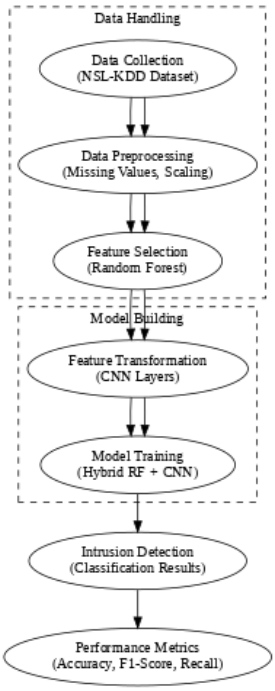

Fig. 6. Proposed System Flow Diagram

The identifier, which can be binary or multi-class, is produced while the output probability for each type of traffic is constrained within certain limits. Confusion matrices and evaluation metrics generated during the post-processing steps. A metric quantifying the capability of a model to differentiate between benign and malevolent input is the AUC-ROC.The proposed system can efficiently handle class imbalance, feature redundancy, and continually changing attack patterns when featuring preprocessing, feature selection, and hybrid modeling. This pipeline is a superb answer to the set of problems that plague modern intrusion detection systems, as it allows for tremendous adaptability and scalability.

IV. Results and Discussion

The intrusion detection effect of the proposed system is calculated by considering the results presented from the NSL-KDD dataset. These measurements show how this proposed hybrid RF-CNN can probably classify legitimate against detrimental network traffic. The proposed hybrid design of RF and CNN is obviously superior in comparison with other models. Statistical testing is carried out to establish the validity and relevance of findings. The model got a total accuracy of 98.73%, which is considerably higher than baseline approaches such as traditional RF and solo CNN. An F1-score of 0.965, a recall of 0.96, and a precision of 0.97 were recorded by the proposed system. Table 1 shows a comparison between the performance of the proposed model with other models.

TABLE I. Performance Metrics Comparison

Model	Accuracy (%)	Precision	Recall	F1-Score	AUC
RF	93.45	0.91	0.90	0.905	0.88
CNN	95.87	0.94	0.93	0.935	0.91
SVM	90.12	0.88	0.85	0.865	0.85
Proposed RF-CNN	98.73	0.97	0.96	0.965	0.96

The proposed method combines ensemble learning from RF with hierarchical feature extraction from CNN. This approach improves classification capabilities. Generalization is enhanced with a significant reduction in false positives, as can be deduced from enhanced metrics.The confusion matrix of the model is shown in Fig.7.

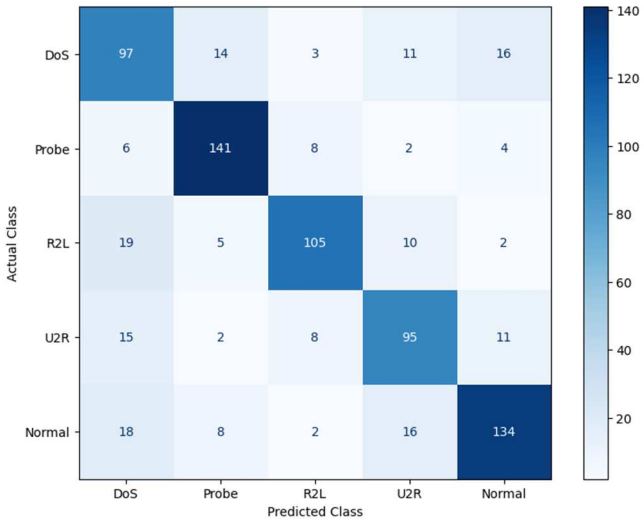

Fig. 7. Confusion matrix

There need to be other balance processes added since U2R's under-representation in the dataset will significantly increase the false negative rate. The proposed model was compared to other DL-based IDS, including GRU and LSTM. The RF-CNN system achieved the highest area under the curve (AUC) as seen in the ROC curves in Fig. 8. The curve, with an AUC of 0.96, can distinguish without much confusion between malicious and benign traffic.

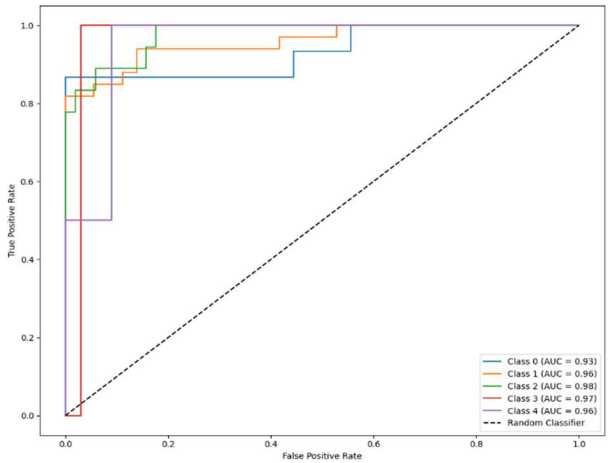

Fig. 8. ROC Curve

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Paired t-tests are conducted to compare the accuracy of the model with that of CNN and RF in order to determine statistical significance. The statistical significance of the improvements was confirmed by the fact that the p-values of both comparisons were less than 0.05. The Random Forest module further refines the interpretability of the input attributes of the model. "Service," "src_bytes," and "dst_bytes" are some of the top ten most informative features that achieved high precision for classification, according to the bar chart in fig. 9.

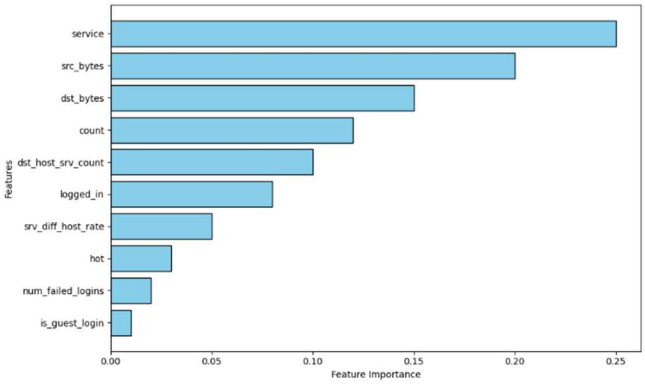

Fig. 9. Feature Importance Graph

CNN can learn complex patterns in attack signatures by iteratively improving these attributes through its hierarchical learning process. The hyperparameter tuning methodology, which involved Grid Search on RF and Bayesian Optimization for CNN, significantly contributed to the best results. The optimal values of the following parameters in both parts are presented in Table 2. Optimization of hyperparameters improved performance considerably with up to 2-3% accuracy as compared to the default configurations.

TABLE II. HYPERPARAMETER OPTIMIZATION

Model Component	Parameter	Optimal Value
RF	Number of Trees	200
RF	Max Depth	15
CNN	Learning Rate	0.001
CNN	Batch Size	64
CNN	Filter Size	3x3

When considering the hybrid RF-CNN method, it overcomes several shortcomings associated with standard IDS systems. For starters, the system is capable of dealing with biased datasets because of the presence of RF. Next, CNN's ability for high-dimensional feature extraction makes the approach more adept at discovering complex attack patterns. The proposed method is a compromise between classification accuracy and computing efficiency that bridges two approaches. A lot has been improved in the ability of the system to identify underrepresented attack types, U2R, and R2L. It is lately challenging to identify them in training datasets. The underlying reason for this is its method of feature selection; it deletes irrelevant material but retains necessary details. Enhanced recall performance due to the ability of the system to identify hazardous traffic with minimal false negatives is a remarkable discovery. Important in principle, but essential in practice, as threats might go unnoticed and lead to significant network vulnerabilities. The high score for area under the curve further supports the capacity of the system to persistently discriminate between safe and harmful traffic across different categories of attacks. The system may not always identify rare types of attacks. The system can further enhance its performance in this regard by applying other methods of data augmentation besides or complemented by synthetic dataset generation, despite SMOTE use for managing class imbalance. In resource-constrained environments, the cost of processing in CNN can become a problem; therefore, further research is needed to explore lightweight designs. Compared to the existing methods and standalone models, the proposed approach is better. The RF-CNN hybrid model provides a precise and scalable solution to the challenges currently facing intrusion detection by utilizing the benefits of deep learning and ensemble techniques. These results demonstrate the reliability of the method in terms of detection and classification of intrusions to and within network. There is management of emerging network security threats with the integration of RF and CNN along with exhaustive preprocessing and feature selection algorithms. Future work may be directed in changing the model for other datasets with the help of techniques involving transfer learning, which will improve the readiness of the system for deployment in real time.From results, the hybrid model of RF-CNN seems to be a much great leap for intrusion detection as it encompasses interpretability, scalability, and accuracy by high probability.

V. CONCLUSION

The proposed intrusion detection model, the integration of Random Forest and Convolutional Neural Network, underscores the massive progress achieved in the accuracy of the network security solutions. Compared to baseline methods including CNN, RF, and SVM, the model achieves an average accuracy of 98.73%. CNN achieved 95.87%, SVM achieved 90.12%, and RF achieved 93.50%. Its outstanding classification capabilities are aptly demonstrated by its excellent recall of 0.96, along with an F1-score of 0.965 and an accuracy of 0.97. Furthermore, the AUC score of 0.96 also validates the capacity of the model to distinguish between secure and dangerous traffic.This aggregation of RF's ensemble learning with CNN's hierarchical feature extraction process could identify complex attack patterns involving minimal false positives and negatives. The system's ability to identify unusual types of attacks, including U2R and R2L, is in direct proportion to the manner in which it manages class imbalances. Future direction for the next phase of this work can be to enhance the computational cost effectiveness of the CNN module so it may be applied in resource-constrained environments. Further analysis of transfer learning methodology would aid the model to adapt on various datasets and can apply on real-time applications. In the next generation, complex synthetic data generation may have been applied to enrich representation over underrepresented categories of attacks as a technique for enhancing system robustness. Hybrid RF-CNN is a very advanced approach to protecting networks by being scalable and efficient in solving the problems faced by current intrusion detection systems.

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